UREA IS NOT EQUALLY DISTRIBUTED BETWEEN THE WATER OF THE BLOOD CELLS AND THAT OF THE PLASMA

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A graphic analysis of some data on the correlation between blood urea concentrations and urea output in man was made by the author some years ago.[1] That analysis led to the conclusion that a cubical parabola was the type of curve that fitted the data best. The curve had an intercept on the blood urea axis at about 9 mg. per cent. While this was suggestive of a threshold, it was known that, with perhaps one exception (1), the idea of a urea threshold had been rather generally rejected (2–5). To us it seemed, however, that at least some of the urea was not freely excreted and existed, perhaps, as a urea complex, or was loosely bound in the blood.

Such "bound urea," whether in the red blood cells or in the plasma, should cause a lack of freedom in the diffusion of urea and an unequal distribution of that substance between the corpuscles and the plasma. This would be contrary to the commonly accepted dicta that urea is freely diffusible and that "urea is about equally divided between the water of the corpuscles and that of the plasma" (6). When the old works (7–16) which were often quoted in support of the equal distribution concept were examined, it was found that, with but one exception (15), no consideration had been given to the difference in the water content of the cells and of the plasma. In that one exception, Ege's view that there is about 80 per cent as much water in the cells as there is in the plasma (17) was used as the basis for the comparison of the urea concentrations of those two blood phases. Using all the old data mentioned above, we recalculated the urea concentrations in the water of the cells and in that of the plasma, making use of the average figures 71 and 93.4 volumes per cent of water in cells and plasma, respectively, and (when cell volumes had not been given) 46 volumes per cent of cells in whole blood. From these new concentrations, cell urea to plasma urea ratios were calculated. Similar ratios were calculated from the data of Boyd (18) who had actually directly determined cell urea and plasma urea but had not been interested in the water concentrations. All the ratios were then plotted in a frequency

[1] Griffith, F. R., Ralls, J. O., and Pucher, G. W., unpublished.

barograph (Fig. 1). It was immediately evident that there was little or no basis for the contention that urea is equally divided between the cell water and the plasma water. Recently, at least one worker (19) has maintained that the ratio of the urea in the *free* water of the cells to that in the *free* water of the plasma varies between 1.06 and 1.96. Thus it seemed to us that the question of the distribution of urea between cells and plasma was by no means settled and we felt justified in continuing our studies along that line. This paper is a report of our findings.

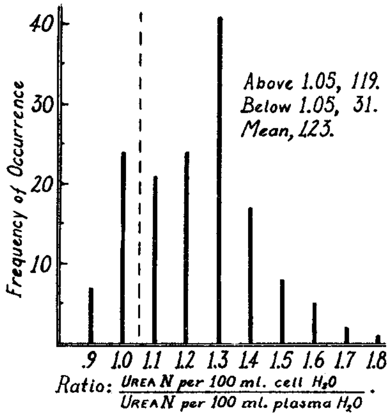

FIG. 1. A barograph showing the frequency of occurrence of various ratios between the concentration of urea nitrogen in the cells and its concentration in the plasma (allowance for the water content of the two phases being made) as calculated from data in the literature.

EXPERIMENTAL

77 samples of blood were contributed by 66 individuals (forty-nine males and seventeen females) within the age range of 20 to 40 years. 66 samples were taken under postabsorptive conditions and eleven were taken under absorptive conditions. The blood was taken by venipuncture without stasis. Heparin was used as the anticoagulant in order to avoid changes in cell volume and any possible shift of urea from the cells to the plasma (18). Plasma was obtained by immediately centrifuging a portion of each blood sample.

Urea nitrogen and total water were determined on the whole blood and the plasma, while cell volume was obtained by hematocrit.

Methods and Errors

Cell Volume—The hematocrit tubes were spun at 3000 R.P.M. for 40 minutes. The results on a single sample of blood were 44.00, 43.96, 44.12, 43.87, 43.85, 43.76, 44.10, 43.99, 43.90, and 43.81, averaging 43.9 ± 0.15, with a maximum spread of 0.3 per cent. In all calculations, the cell volume was rounded to the nearest 0.1 per cent.

Total Water—A drying and weighing method was adopted after that of Gurevitch and Karlson (20) had been found to be unsatisfactory. A temperature of 150° and a drying time of 17 hours were selected, although Nelson and Hulett (21) recommended 325°, the critical temperature of water. The samples were measured by means of an Ostwald-Van Slyke pipette (delivering 1.007 ml.) and the temperature of the blood or plasma was noted. After the material was dried, the weight loss was divided by the density of water at the temperature noted. The quotient was corrected by the factor 99.3 (1/1.007 × 100) to give the per cent water in the sample. The results on a single sample of whole blood were 83.3, 83.6, 83.8, 83.9, 83.5, and 83.7 per cent, with a maximum deviation of 0.6 per cent. The results on a single sample of plasma averaged 93.4 per cent, with a maximum spread of 0.3 per cent.

Urea Nitrogen—This quantity was determined by the manometric method of Van Slyke (22, 23), but with certain modifications described below.

Because difficulty was experienced in obtaining sufficiently accurate pressure readings at the 0.5 ml. mark, the practice of measuring all pressures at the 2.0 ml. graduation was adopted. This practice eliminated the major part of the error introduced when a small bit of stop-cock grease got into the capillary just below the upper stop-cock. A 0.01 ml. piece of grease at this point introduces an error of 2 per cent when the pressures are read at 0.5 ml., whereas, when they are read at the 2.0 ml. mark, just a 0.5 per cent error is introduced. As an example of the troubles thus avoided, the following pressure readings on the same quantity of trapped air are given, with the usual 2 minute extraction period preceding each reading: at 0.5 ml. and 28.5°, 31.64, 31.97, 31.80, 31.98, 31.78, 31.98, 31.65, 32.02, 31.89, 31.95, 31.73, 31.70, 31.80, 32.02, 31.95, 31.78, 31.92, and 31.98, maximum deviation 0.38 cm.; at 2.0 ml. and 28.5°, 13.12, 13.13, 13.12, 13.14, 13.12, 13.14, 13.12, 13.14, 13.14, 13.14, 13.12, 13.12, 13.11, 13.14, 13.14, and 13.13, maximum deviation 0.03 cm. Thus it was evident that, although the urea nitrogen factor per mm. of carbon dioxide pressure at 2.0 ml. is approximately 4 times that at 0.5 ml., a considerable net gain in accuracy was effected.

It was found necessary to run frequent blanks during each day, because these invariably increased as the day progressed. Howell (24) also found that changes in the blank occurred as the age of the urease solution increased.

In practice, the time at which the urease solution was added to the reaction chamber was noted in all blank and experimental determinations. The blanks for experimental determinations were obtained by interpolation between the preceding and succeeding blanks. The urease solutions were prepared from Squibb's Double Strength urease powder.

Correction Factors—The reaction chamber was calibrated at the 2.0 ml. mark and was found to have an actual volume of 1.988 ml. This introduced a correction factor of 0.994, which in conjunction with that of 0.993 for the pipette gave an over-all factor of 0.986. All our determinations of urea nitrogen were corrected by means of this last factor.

Check of Procedure and Technique—For this purpose, we used a solution containing 15 mg. of urea nitrogen and 0.8437 gm. of sodium bicarbonate (equivalent to 45 volumes per cent of carbon dioxide) per 100 ml. The following results were obtained: 14.85, 14.90, 15.10, 14.99, 15.08, 15.02, 15.02, 14.89, 14.91, 15.10, 15.11, 14.85, 14.93, 15.02, 15.00, 15.00, 15.15, 14.85, and 14.86, with an average of 14.98 mg. of urea nitrogen per 100 ml. The maximum range was ± 1 per cent of the correct value.

Effect of Each Error on Ratio of Urea N per 100 Ml. Cell Water to Urea N per 100 Ml. Plasma Water—An absolute error of ± 0.3 per cent in the cell volume introduces a negligible error of ± 0.005 in the calculated ratio, because the same proportionate error is reflected in both the cell urea nitrogen and the cell water calculations.

An absolute error of ± 0.6 per cent in the water content of whole blood introduces an error of ± 0.02 in the ratio, while one of ± 0.3 per cent in the plasma water introduces an error of ± 0.005 in the ratio of cell urea to plasma urea.

A relative ± 1 per cent error in the urea nitrogen of whole blood produces an error of ± 0.03, whereas a similar error in the plasma urea nitrogen figure means an error of ± 0.025 in the ratio.

The most adverse combinations of the above errors (high whole blood urea, high plasma water, low plasma urea, and low whole blood water, or *vice versa*) introduce an error of ± 0.08 in the ratio of the concentration of urea nitrogen per 100 ml. of cell water to the urea nitrogen per 100 ml. of plasma water.

Influence of Arginase Activity on Urea Nitrogen Determinations—Although it was felt that Howell (24) had adequately shown that the "arginase error" in urea determinations resided in an enzyme-substrate system in the urease preparation itself and that we had eliminated that through our use of frequent blank determinations, it seemed advisable, in order to preclude any adverse criticism, to show, if possible, that this "error" was inoperative in our determinations. Accordingly, a study was made of the action of the usual 50 per cent glycerol extract of urease (23) upon arginine carbonate in water and upon blood to which arginine carbonate had been

TABLE I

Evidence That Action of Arginase Was Insignificant in Our Determination of Distribution of Urea between Human Erythrocytes and Plasma

Sample No.	Description	pH of sample*			pH of reaction mixture*			Urea nitrogen, mg. per 100 ml.			Ratio†		
		1 hr.‡	24 hrs.	48 hrs.	1 hr.	24 hrs.	48 hrs.	1 hr.	24 hrs.	48 hrs.	1 hr.	24 hrs.	48 hrs.
1	Aqueous§ arginine carbonate				6.90	6.90		0.00	0.00	0.00			
2	" "				6.91	6.90		0.00	0.00	0.00			
3	" "				6.95	6.94	6.95	0.00	0.00	0.00			
4	Whole blood	7.28	7.36		6.95	7.03		14.08	13.70		1.12	1.12	
4	Plasma from Sample 4	7.29	7.35		6.90	6.95		15.24	14.85				
4a	Whole blood, Sample 4, + arginine carbonate	7.27	7.37	7.36	7.01	7.01	7.02	14.08	15.60	16.70	1.12	1.12	1.12
4a	Plasma from Sample 4a	7.26	7.33	7.35	6.96	6.95	7.00	15.24	17.00	18.20			
5	Whole blood	7.30	7.33		6.92	6.93		9.48	9.03				
5a	" " Sample 5, + arginine carbonate	7.28	7.34		6.90	6.91		9.43	11.33				
6	Whole blood	7.30	7.36	7.32	7.01	7.02	7.02	13.76	13.34		1.12	1.12	
6	Plasma from Sample 6	7.27	7.33		6.97	6.98		14.75	14.31				
6a	Whole blood, Sample 6, + arginine carbonate	7.31	7.40	7.35	6.96	7.00	7.02	13.73	15.25	16.36			1.12
6a	Plasma from Sample 6a			7.30			7.01			17.50			

* The pH determinations were made by means of a quinhydrone electrode with a type K potentiometer and a glass electrode with a Beckman meter. The values obtained for buffers made to have pH values of 4.00, 6.6, and 7.2 were 4.01, 6.57, and 7.23, respectively. In addition to the pH of the reaction mixtures shown in the table, another whole blood with and without added arginine was tested. The results were 7.00 and 6.92, respectively.

† This ratio is that between the cell urea nitrogen per 100 ml. of cell water and the plasma urea nitrogen per 100 ml. of plasma water.

‡ These times represent the maximum interval from the blood letting and addition of arginine carbonate (or the preparation of the arginine carbonate solution) to the time when the indicated determinations were made.

§ The arginine carbonate solution was made to contain 120 mg. per 100 ml. Analyses showed it contained 32.0 mg. of nitrogen per 100 ml., of which approximately 16 mg. were available for conversion to urea nitrogen by arginase. Arginine carbonate was added to blood to give the same concentration as the above solution. Thus, if the arginase was significantly active, it would have been possible to get an extra 16.00 mg. of urea nitrogen in our blood analyses.

added as compared with its action upon a portion of the same blood to which no arginine had been added. These studies were made under conditions exactly analogous to those prevailing in our urea nitrogen determinations as reported herein. The pH of the reaction mixture in the Van Slyke manometric determination of urea nitrogen, as applied to the above arginine carbonate solutions and the blood with and without added arginine, was determined. The results (Table I) indicate that, under the conditions of our regular determinations, no extra urea was derived from an "arginase action" upon arginine in the blood.

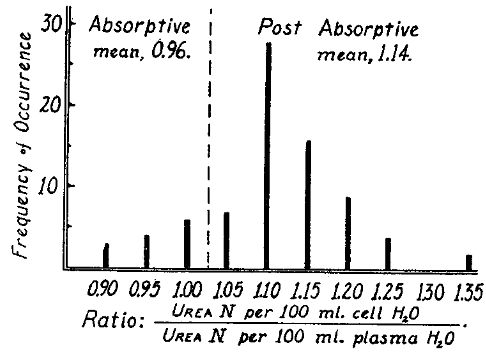

FIG. 2. A barograph showing the frequency of occurrence of various ratios between the concentration of urea nitrogen in the water of red blood cells and its concentration in the plasma water, as determined on the blood of 66 individuals in the postabsorptive state and thirteen in the absorptive state.

Results

The results of the distribution studies are given in two forms: (1) in a barograph (Fig. 2) depicting the frequency of occurrence of the observed ratios, rounded to the nearest 0.05; and (2) in Table II in which only the averages of the results obtained are given for certain groups of blood: Group A, male and female, 66 postabsorptive samples, Group B, female, seventeen postabsorptive, Group C, male, forty-nine postabsorptive, Group D, male, thirteen postabsorptive, and Group E, eleven samples from the same individuals as in Group D, but in the absorptive state. The ranges, standard deviations, standard deviations of the means, and the probable errors of the means for each group are given.

DISCUSSION

The average of all the ratios between the concentration of the urea in the total water of the cells and that in the total water of the plasma was 1.14 $\pm$ σ_M = 0.007. None of the individual postabsorptive ratios was less than 1.04, while one was as high as 1.39. If it was assumed that each ratio was 0.08 too high (the result of a combination of the following errors: a relative +1 per cent in the whole blood urea, a relative −1 per cent in the plasma urea, an absolute +0.3 per cent in the plasma water, and an absolute −0.6 per cent in the whole blood water) and each ratio was corrected for this, the average ratio would still be 1.06, or more than 1. Such a combination of errors could not have been operating in the analyses of postabsorptive blood in all 66 samples, nor is it probable that it was operating in all sixteen of those analyses that resulted in ratios as low as 1.04 to 1.08.

The author is well aware of the work of Behre (25), Addis (26), and Anderson and Tompsett (27) who pointed to arginase, either in the blood cells or in the urease preparations, as a source of error in blood urea determinations. Moreover, Edlbacher, Krause, and Merz (28) definitely proved the existence of arginase in blood, while Sumner and Dounce (29), Howell (24), and Hellerman *et al.* (30–32) have certainly shown that an arginolytic enzyme is present in jack bean urease preparations. However, Van Slyke (23), in describing his procedure for blood urea nitrogen, stated that the arginase error was not a factor, while Howell (24), having shown that the error was due to an enzyme-substrate system residing in the urease preparations, suggested means (which we adopted in part) of eliminating it. We are certain that no arginase error was present in our results, because the urease preparation which we used did not demonstrably act upon arginine carbonate during the usual incubation period of a urea nitrogen determination, nor did it liberate any detectable quantity of "extra" urea from blood to which arginine had been added (Table I). This was not surprising in view of the fact that the pH of the reaction mixtures in our urea nitrogen determinations fell within the limits 6.90 to 7.02 and the fact that the pH range of activity of arginase is 7 to 11 (24, 30–34) with an optimum at pH 9.5, or, in the presence of cobaltous ions, at pH 7.5 (31).

According to calculations based upon the various reported velocity constants for the hydrolysis of arginine by arginase at pH 7.5 with added cobaltous ions (30, 31), only 1.2 to 6.5 per cent of any arginine present would be hydrolyzed in the first 3 minutes after the addition of the enzyme. But, under the conditions of our determinations (no added cobalt, a pH of 7.0 instead of 7.5, and 1 to 2 minutes of incubation instead of 3), even less than that fraction of any arginine present would have been hydrolyzed to urea. Even if one assumed that the conditions were ideal for the optimal activity of the arginolytic activity of the jack bean urease preparations and that of the approximately 12.5 mg. of amino acid nitrogen in the red blood cells (35) about 1.25 mg., a high estimate (36, 37), may have been arginine α-amino nitrogen, there could have been, at most, but

2.5 × 0.06, or 0.15 mg., of "extra" urea nitrogen from that source in our cell urea figures. This amount would have been insufficient to invalidate our calculations of the ratio of cell urea nitrogen to plasma urea nitrogen.

Hence, in view of all the above considerations, it appears certain that urea is not equally divided between the water of the red blood cells and

TABLE II

Average Analytical Figures, Ratios, and Statistical Data on Blood of Different Groups of Males and Females in Postabsorptive and Absorptive State

Group	Whole blood			Plasma			Cells			Ratio, M (b/a)	Statistical data on ratios
	Urea N / 100 ml.	Per cent water	Per cent cells	Urea N / 100 ml.	Per cent water	Urea N / 100 ml. water (a)	Urea N / 100 ml.	Per cent water	Urea N / 100 ml. (b)		
	mg.			*mg.*		*mg.*	*mg.*		*mg.*		
A	12.63	83.4	45.7	13.46	93.5	14.37	11.65	71.4	16.31	1.14	R = 1.04–1.39 σ = ±0.065 σ_M = ±0.008 E_M = ±0.005
B	13.00	85.0	41.9	13.65	93.7	14.55	11.96	72.7	16.40	1.13	R = 1.05–1.24 σ = ±0.061 σ_M = ±0.015 E_M = ±0.01
C	12.50	82.9	47.0	13.39	93.4	14.31	11.54	70.9	16.28	1.14	R = 1.04–1.39 σ = ±0.066 σ_M = ±0.009 E_M = ±0.006
D	12.80	83.4	45.8	13.75	93.3	14.71	11.73	71.5	16.28	1.12	R = 1.08–1.16 σ = ±0.024 σ_M = ±0.008 E_M = ±0.005
E	13.06	83.6	45.8	14.81	93.4	15.87	10.99	71.8	15.27	0.96	R = 0.90–1.01 σ = ±0.034 σ_M = ±0.01 E_M = ±0.007

Group A, 66 males and females in the postabsorptive state; Group B, seventeen females, postabsorptive; Group C, forty-nine males, postabsorptive; Group D, thirteen males, postabsorptive; and Group E, eleven males from Group D in the absorptive state. M, mean; σ, standard deviation; σ_M, standard deviation of the mean; E_M, probable error of the mean. $(M_D - M_E)/(E_{M_D - M_E}) = 20$, the ratio of the difference between means D and E to the probable error of that difference.

that of the plasma. In fact, as stated above, we found the average ratio of cell urea per 100 ml. of cell water to plasma urea per 100 ml. of plasma water to be 1.14. It was a matter of further interest to note that there is no significant difference between the distribution of urea in the blood of females and that in the blood of males (Table II).

Although it may be admitted that our data establish that urea is not equally distributed between the total water of the cells and that of the plasma, some might argue that it might be equally divided between the free water of those two blood phases. But, Stolzman (19) has reported evidence to the contrary. He reported ratios varying between 1.06 and 1.96 after having calculated the free water in the cells by means of the relative proportions of chloride in the cells and in the plasma and the assumption that that ion is equally distributed between the free water of those two parts of the blood. However, his calculations are subject to considerable doubt, for others (38–40) have shown that chloride is not so distributed. While we have made no direct determinations of the free water in the bloods analyzed, we have assembled the ratios calculated from

TABLE III

Ratios of Distribution of Urea between Corpuscles and Plasma, with Total Water and Free Water Content As Bases of Calculation

Ratio	Basis of calculation
1.14	Our own data on urea and total water
1.13	" " " " " " water, and average protein content of cells and plasma, and degrees of hydration: 0.14 ml. per gm. hemoglobin (41); 0.11 ml. per gm. albumin (42); 1.87 ml. per gm. fibrin (43); and 1.75 ml. per gm. globulin
1.19	Our own data on urea and total water content, and free water as calculated from degrees of hydration of hemoglobin and albumin given above and degrees of hydration of fibrinogen and globulin (per gm.) assumed the same as for albumin

available data for free water and urea in Table III. It is evident that, at least with the information now at hand, the ratio cannot be made to be the 1:1 that the equal distribution concept demands.

The distribution of urea in blood collected from 1 to 3 hours after breakfast was found to be very different from that in blood from individuals in the postabsorptive state. The average ratio was 0.96 (Group E, Table II). This was, undoubtedly, a reflection of the reported rise in blood urea following a meal (44–47). Folin and Svedberg (15) published data which showed that a preferential increase in the plasma urea occurred after a meal. However, they did not happen to call attention to it. When we took their average figures for urea nitrogen in whole blood and in plasma and, using our average cell volumes per cent and volumes per cent of water in whole blood and in plasma, calculated distribution ratios, we obtained 1.06 and 0.92 for postabsorptive and absorptive samples, respectively. In regard to our own data, it may be said that, although one should not apply statistical methods to as few as eleven results, such an application here indicates that the chances are better than 1 billion to 1 that the difference between the absorptive and postabsorptive ratios in blood is real. This difference is not unlike, except in magnitude, the difference between the distribution of lactic acid in human blood during glycolysis and that in blood drawn in the basal state (48). During glycolysis, the lactic acid accumulated more rapidly in the plasma than it did in the cells. This observation and ours for urea could, at least in part, be explained as being due to a slow diffusion through the cell membrane. In fact, we found that, when plasma was replaced by mammalian Ringer's solution (containing no urea), the urea nitrogen in the external fluid did not reach a maximum until about 1 hour afterwards. The rate of diffusion of urea into and out of red blood cells is to be more extensively investigated.

The exact cause of the demonstrated accumulation of urea within the red blood cells (in the postabsorptive state) is not known. The polymerization and tautomerization of urea, the rapid removal of urea from plasma at the kidney, the metabolic production of urea within the erythrocytes, and the intracellular fixation of urea by adsorption on or its chemical union with some cell constituent were all considered. The polymerization and tautomerization were discarded as untenable after some consideration. The possibility that a rapid elimination of urea from the plasma and a slow diffusion from the cells might account for the unequal distribution observed was discounted by the observation that, under conditions which permitted decreases and increases in the urea of different portions of the same blood sample, the distribution remained the same as it was in the freshly drawn blood (Table I). Nor was it considered likely that a metabolic production of urea within the erythrocytes was the explanation. Such an action could lead to a constant absolute difference between the urea concentration in the cells and that in the plasma (a steady state in compliance with Frick's diffusion law), but high distribution ratios should then accompany low urea concentrations and *vice versa*. There was no such relation between the ratios and the urea values.

Two equations relating the distribution of urea between the cells and plasma to certain known and certain assumed factors were developed from Langmuir's adsorption isotherm, $(1 - \Theta)\alpha\mu = \nu\Theta$. The one, for appreciable adsorption, appeared to be not applicable, for it carried a term involving the urea concentration and it had been observed that the distribution was not related to the total urea present. However, that for but slight, or poor, adsorption reduced, in its simplest form, to $R = 1 + (km/H_2O)$, where R is the distribution ratio, k is a constant, m is the number of gm. of adsorbent in 100 ml. of cells, and H_2O is the volumes per cent of water in the cells. In developing this equation it was assumed that there was a portion of the urea within the cells in osmotic equilibrium with the urea in the plasma and that there was another portion adsorbed on some material. Curiously, a very similar equation was derived from the mass action law and the assumption that some cell urea reversibly combined with some cell constituent in a bimolecular reaction. That equation was $R = 1 + (kM/H_2O)$, in which, as before, the symbols had the same significance, except that M represents the number of moles of non-urea reactant per 100 ml. of cells. Since neither of these equations involved the actual concentration of urea and since our ratios were not related to that actual concentration, it seemed that the underlying assumptions, that some urea was relatively fixed within the cell either by adsorption or by chemical union with a cell solute, were the most probable explanations of the unequal distribution of urea between the water of the cells and that of the plasma. Unfortunately, the analytical data afforded no means of deciding between those two possibilities.

Of all the cell constituents that might form a non-diffusing urea complex, hemoglobin is the most likely. The effect of urea on hemoglobin (49–53) and the behavior of heme with nitrogenous organic compounds (54–60) are well known. While very high concentrations of urea are necessary to produce a demonstrable denaturation of hemoglobin, it is possible that similar changes, to a lesser degree, might also occur at low urea concentrations. Moreover, it is possible that (though no claim is made that it does occur) urea might compete with globin for heme. Either action could easily account for our postabsorptive ratios, for, to do so, each gm. of hemoglobin would need to adsorb but 0.04 mg. of urea, or but 1 of every 23 molecules of hemoglobin (or 1 out of every 92 molecules of heme) would need to combine with just 1 of every 10 molecules of urea within the cells. In view of the above possibilities, it is our intention to investigate the "urea-binding" potentialities of hemoglobin and of heme.

We wish to thank E. R. Squibb and Sons for a generous supply of Squibb's Double Strength urease.

SUMMARY

1. An "arginase error" was not responsible for the results reported herein.

2. 66 blood samples from individuals in the postabsorptive state were analyzed. The individual and average figures are given for the urea nitrogen and the total water in the whole blood, in the plasma, and in the cells.

3. Urea is not equally divided between the water of the cells and that of the plasma. The distribution ratio is 1.14:1.00, on the average.

4. During the active production of urea after a meal, the ratio tends to be less than 1:1.

5. The possible factors involved in the unequal distribution of urea are discussed.

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